

Jingjing Pan

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Birthday: 25.8.1995 Nationality: China

POSITION	Postdoc Fellow, Karlsruhe Institute of Technology	2025.10.1 - present
	Research Associate, Lawrence Berkeley National Lab	2022.5.1 - 2025.8.31
EDUCATION	Ph.D. in Physics, Highest Honors, Yale University ,	2018.9.1 - 2024.12.19
	• Dissertation: <i>Exploring the Standard Model and Beyond through the Lens of Jet Substructure and Deep Learning with the ATLAS Experiment</i> • Advisory Committee: Profs. Keith Baker, Helen Caines, Sarah Demers, Ian Moulton, Benjamin Nachman	
	B.S. in Physics, Math, Penn State University	2015.9.1 - 2018.8.21
	• GPA: 3.95/4.00 • Advisor: Prof. Mikael Rechtsman	
RESEARCH	The ATLAS Experiment at the LHC	2019 Dec. - 2025 July
	• First measurement of track functions (paper), the Contact • Deep learning methodology for data analysis - New deep learning-based unfolding method (paper), the Contact	
	The H1 Experiment at HERA	2023 June - present
	• First <i>fully differential</i> measurement (proceeding) - First measurement of all particles in the H1 data <i>beyond the lab frame</i>	
AWARDS / GRANTS	• Co-PI of project of 5.5M CPU & 87k GPU hours at supercomputer HoreKa (145,784 € equivalent) 2026 • Young Investigator Group Prep Program Fellowship (100,000 €, KIT) 2026 • Alexander von Humboldt Postdoctoral Fellowship 2026 • Contributor (7/9 pages) of the Yale High Energy Group U.S. Department of Energy proposal (funded 2 PhD at Yale, 2023-26 fiscal year) 2022 • U.S. ATLAS ATC Research Award (\$11,000 for visit at LBNL) 2021 • Lead Teaching Fellowship (Yale) 2019	

- Yale Early Start Research Grant (\$10,000) 2018
- Downsborough Chair in Physics Research Award (\$7,000) 2018
- Department of Physics Teaching Award (1st undergrad recipient) 2018
- Bert Elsbach Scholarship (for academic performance) 2017
- Dean's List 2015-2018

INVITED / PLENARY TALKS

- Overview on neural simulation-based inference (SBI) for particle physics
([slides](#)) AI@LHC, 2026.5.26
- Measurement of jet track functions using deep learning with the ATLAS experiment
([talk](#)) ML4Jets, Paris, 2024.11.7
- Jet substructure measurements in multijet production with the ATLAS experiment
([talk](#)) BOOST, Genoa, 2024.7.30
- Enabling Specialized Unfolding Methods with Modern Deep Learning Techniques
([talk](#)) France-Berkeley PHYSTAT Workshop, Paris, 2024.6.11
- Probing higher order DGLAP evolution using OmniFold and Energy Correlators
([talk](#)) BOOST, Berkeley, 2023.8.3
- Improving and Extending Precision Jet Substructure with Modern Deep Learning Techniques
LBNL HEP Seminar, 2024.6
- Overview of Jet Calibration Techniques on ATLAS
(session co-chair) Hadronic Calibration Workshop, 2021.8

TEACHING

Yale University

- Mathematical Methods Fall 2022
- Lecture on Group Theory ([notes](#))
- Nuclear and Particle Physics Spring 2022
- Lecture on research in ML techniques
- Classical Mechanics Fall 2021
- Lecture on Path Integrals ([notes](#))
- Introductory Physics Lab Fall 2018-Spring 2020
- Lead Teaching Fellow
- Designed exam rubrics, organized grading, facilitated exam reviews and lab sessions, graded reports

Pennsylvania State University

- Statistical Mechanics Spring 2018
- Led recitations and reviews, helped with assignments

- Introductory Modern Physics Spring 2018
- Helped with assignments and held reviews
- Theoretical Mechanics Fall 2017
- Wrote psets and led in-class practice sessions

MENTORSHIP **Yale Univeristy**

- Nathan Suri (PhD, formerly at Caltech) 2022.5-2024.5
- *ML Pileup Mitigation for HL-LHC*
- Neal Ma (Bachelor in Physics & Computer Science) 2022.4-2023.4
- *Optimizing step2 of OmniFold*
- Vinicius Da Silva (Bachelor, now at Fermilab) 2020.10-2021.7
- *Search for $H \rightarrow Za \rightarrow 4l$ ($l = e, \mu$) in ATLAS Run2 data*
- Asmaa Aboulhorma (PhD in Morocco, $H \rightarrow ZZ_d$) 2021.9-2022.5
- Zainab Soumaimi (PhD in Morocco, $H \rightarrow ZZ_d$) 2021.6-2022.1

SKILLS

Coding: Python, C++, MATLAB, Scheme (functional)

Tools: GNU/Linux, ROOT, NumPy, Uproot/Awkward, pandas, TensorFlow, Keras, PyTorch (basic), Matplotlib, L^AT_EX, Delphes, Mathematica, Horovod, COMSOL, [MPB](#), [MEEP](#)

Languages: English (proficient), Mandarin (native), Japanese (intermediate), Cantonese (intermediate)

OUTREACH/ SERVICE

- Reviewer, [Neural Information Processing Systems \(NeurIPS\) ML4PS Workshop](#) (2024)
- Reviewer, SciPost, Proceedings A
- Moderator, [Kimball Smith Series](#) (Yale Jackson School of Global Affairs, 2023)
- Organizer, ATLAS Group Journal Club (Yale, 2022-2024)
- TA, [US ATLAS Machine Learning Training](#) (LBNL, 2023)
- Volunteer, US ATLAS Summer Workshop (Yale, 2023)
- Mentor, Women in Physics (Yale, 2019-2022)
- Member, Yale Physics Climate & Diversity Committee
- Organizer, Conference for Undergraduate Women in Physics ([Yale](#), 2020)
- Participant, Conference for Undergraduate Women in Physics (Princeton, 2017; Columbia, 2018)
- Volunteer, Science Booth Outreach at central PA Arts Fest (2017)